

Homework 6

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Problem 9.3

Problem 1.1. *Compute the flow of the vector field*

$$V = y \frac{\partial}{\partial x} + \frac{\partial}{\partial y}.$$

Proof. The flow $\theta : \mathbb{R} \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$ must be such that the curve $\gamma := \theta(-, p) = \theta_t(p)$ satisfies $\gamma'(t) = V(\gamma(t))$. We then have in coordinates

$$\frac{d\gamma_x}{dt} \frac{\partial}{\partial x} + \frac{d\gamma_y}{dt} \frac{\partial}{\partial y} = \gamma_y(t) \frac{\partial}{\partial x} + \frac{\partial}{\partial y}.$$

This gives us the system

$$\frac{d\gamma_x}{dt} = \gamma_y, \quad \frac{d\gamma_y}{dt} = 1.$$

We solve the system by

$$t + C = \int \frac{d\gamma_y}{dt} dt = \int d\gamma_y = \gamma_y.$$

Then

$$\gamma_x(t) = \int \frac{d\gamma_x}{dt} dt = \int \frac{d\gamma_y}{dt} dt = \int t + C dt = \frac{1}{2}t^2 + Ct = D.$$

The solution must satisfy the initial condition $\gamma(0) = p$. Write $p = (x, y)$. This yields

$$\theta_t(x, y) = \left(\frac{1}{2}t^2 + yt + x, t + y \right).$$

■

Problem 11.5

Problem 2.1. *Given M a manifold, show T^*M is a trivial vector bundle if and only if TM is trivial.*

Remark 2.1. A (real) vector bundle of rank k over M is a space E and a continuous surjection $\pi : E \rightarrow M$ such that

1. for each $p \in M$, the fiber $E_p := \pi^{-1}(p)$ is a $\mathbb{R}$ -vector space of dimension k ;
2. for each $p \in M$, there is an open neighborhood $p \in U \subseteq M$ and a homeomorphism $\Phi : \pi^{-1}(U) \cong U \times \mathbb{R}^k$ called the *local trivialization of E over U* , satisfying the following conditions:
 - If $\pi_U : U \times \mathbb{R}^k \rightarrow U$ is the projection, then $\pi_U \circ \Phi = \pi$;
 - For each $q \in U$, the restriction $\Phi|_{E_q}$ is a vector space isomorphism $E_q \cong \{q\} \times \mathbb{R}^k \cong \mathbb{R}^k$.

If M, E are smooth manifolds, π is smooth, and the local trivializations can be chosen to be diffeomorphisms, then E is a *smooth* vector bundle of rank k over M . If there exists a local trivialization of E over all of M , called a global trivialization, then E is called a *trivial* vector bundle.

Proof. Throughout the problem, let e^i be the i th basis vector in the standard basis for $\mathbb{R}^n$

$$e^i = (0, \dots, 0, 1, 0, \dots, 0)$$

with a 1 in the i th spot.

($\Leftarrow$) Let TM be trivial with $\Phi : TM \cong M \times \mathbb{R}^n$ the global trivialization. Write the component functions as $\Phi(p, v) = (\phi'(p), \phi(v))$. Let $\pi_M : M \times \mathbb{R}^n \rightarrow M$ and $\pi : TM \rightarrow M$ be the natural projections. Since $\pi_M \circ \Phi = \pi$, we have

$$p = \pi(p, v) = (\pi_M \circ \Phi)(p, v) = \pi_M(\phi'(p), \phi(v)) = \phi'(p).$$

Therefore, ϕ' is the identity, meaning we can write $\Phi(p, v) = (p, \phi(v))$.

Write Φ_p for the restriction $\Phi|_{T_p M} : T_p M \cong \{p\} \times \mathbb{R}^n$. It's important to note here that when viewed as a subset of TM , the space $T_p M$ takes elements of the form (p, v) with v tangent to p . This means we should really be writing $\{p\} \times T_p M$, but this is annoying and the spaces are isomorphic so we will abuse notation throughout this problem in this way. We will similarly write $T_p^* M$ for the subspace $\{p\} \times T_p^* M \subseteq T^* M$. This means that we write $\Phi_p(p, v) = (p, \phi(v))$.

With this notation established, we can begin the problem. Since Φ is a trivialization, each Φ_p is an isomorphism of vector spaces, meaning there are inverses $\Phi_p^{-1} : \{p\} \times \mathbb{R}^n \rightarrow T_p M$. Since isomorphisms of vector spaces map bases to bases, we note that

$$\{V_p^i := \Phi_p^{-1}(e^i)\}_{i=1}^n$$

is a basis for $T_p M$. To define a covector, we need only specify its action on basis elements, so we can define a covector $\omega_p^i \in T_p^* M$ for each p and for each $1 \leq i \leq n$ which acts as

$$\omega_p^i(V_p^j) = \delta_i^j.$$

First, we claim that the set $\{\omega_p^i\}_{i=1}^n$ is a basis for $T_p^* M$.

Since $T_p^* M$ is n -dimensional, it suffices to show that $\{\omega_p^i\}_{i=1}^n$ is a linearly independent set. Let there exist coefficients c_i for $1 \leq i \leq n$ such that

$$\sum_{1 \leq i \leq n} c_i \omega_p^i = 0.$$

Then this covector must map all vectors to 0; in particular,

$$0 = \left(\sum_{1 \leq i \leq n} c_i \omega_p^i \right) (V_p^j) = \sum_{1 \leq i \leq n} c_i \omega_p^i(V_p^j) = \sum_{1 \leq i \leq n} c_i \delta_i^j = c_j.$$

Therefore, each c_i is 0, so the ω_p^i s are linearly independent.

We now claim that for each i , the ω_p^i s assemble into a smooth covector field $\omega^i \in \mathfrak{X}^*(M)$. We also claim that the collection $\{\omega^i\}_{1 \leq i \leq n}$ is a smooth global frame for $T^* M$ in the sense of the following definition. By Lee 10.20, a smooth vector bundle is trivial if and only if it admits a smooth global frame, so showing this suffices for this direction of the proof. We must show the following is satisfied.

Definition 2.2 (Lee page 257). Let $E \rightarrow M$ be a vector bundle. If $U \subseteq M$ is an open subset, a k -tuple of local sections $(\sigma_1, \dots, \sigma_k)$ of E over U is said to be *linearly independent* if their values $(\sigma_1(p), \dots, \sigma_k(p))$ form a linearly independent k -tuple in E_p for each $p \in U$. A *local frame* for E over U is an ordered k -tuple $(\sigma_1, \dots, \sigma_k)$ of linearly independent local sections over U that span E ; thus $(\sigma_1(p), \dots, \sigma_k(p))$ is a basis for the fiber E_p for each $p \in U$. It is called a *global frame* if $U = M$. If $E \rightarrow M$ is a smooth vector bundle, a local or global frame is a *smooth frame* if each σ_i is a smooth section.

We have already shown that the tuple $(\omega^1, \dots, \omega^n)$ is a global frame since they are sections of T^*M by definition, and their evaluations form bases as explored earlier. It suffices to show they are smooth. By Lee 11.11, it suffices to show for any $X \in \mathfrak{X}(M)$, the map $\omega^i(X)$ is smooth.

First, note that V^i is a smooth vector field since by definition, $V_p^i = \Phi_p^{-1}(e^i)$, and Φ^{-1} is smooth. Since the collection of V^i 's are linearly independent in the above sense, they form a global frame for TM . By the basis condition, given a vector field $X \in \mathfrak{X}(M)$ we can write

$$X_p = \sum_i c_i(p) V_p^i$$

for some coefficients $\{c_i(p)\}_{1 \leq i \leq n}$. We can assemble these into functions $c_i : M \rightarrow \mathbb{R}$ such that

$$X = \sum_i c_i(p) V^i.$$

We would like to show the c_i 's are smooth. Applying Φ yields

$$\Phi(X_p) = \Phi_p \left(\sum_i c_i(p) V_p^i \right) = \left(p, \sum_i c_i(p) \Phi_p(V_p^i) \right) = \left(p, \sum_i c_i(p) e^i \right).$$

Now let $\pi_{\mathbb{R}} : M \times \mathbb{R} \rightarrow \mathbb{R}$ be the projection $(p, x) \mapsto x$. Since $X, \Phi, \pi_{\mathbb{R}}$ are all smooth, the composite $\pi_{\mathbb{R}} \circ \Phi \circ X$ is smooth. To determine what this composite is, we follow an element $p \in M$:

$$p \xrightarrow{X} X_p = \sum_i c_i(p) V_p^i \xrightarrow{\Phi} \left(p, \sum_i c_i(p) e^i \right) \xrightarrow{\pi_{\mathbb{R}}} \sum_i c_i(p) e^i.$$

Note that the component functions of this map are the c_i 's, and since a map is smooth if and only if its component functions are smooth, it follows that c_i is smooth for all i .

Now we simply apply ω^i to X :

$$\omega^i(X)_p = \omega_p^i \left(\sum_j c_j(p) V_p^j \right) = \sum_j c_j(p) \delta_j^i = c_i(p).$$

Therefore, $\omega^i(X) = c_i$. Since c_i is smooth, $\omega^i(X)$ is smooth. By Lee 11.11, ω^i is smooth, and thus forms a smooth global frame, completing this direction by 10.20.

($\implies$) The proof for this direction is exactly the same so we offer less details. Let $\Phi : T^*M \rightarrow M \times \mathbb{R}$ be a global trivialization with notation as before. Define ω_p^i as $\Phi_p^{-1}(e^i)$. There exists no natural isomorphism $T^*M \cong (T^*M)^*$, but there is a natural isomorphism $TM \cong (T^*M)^*$ by basic linear algebra. We utilize this fact as follows. Define $V_p^i \in T_p M$ as the vector $W_p^i(\omega_p^j) = \delta_j^i$; since the ω_p^j 's are a preimage of a basis and thus a basis, this suffices to specify W_p^i . Then define V_p^i to be the

element of TM corresponding to W_p^i under the natural isomorphism. Recall the definition of this isomorphism: for any $v \in V$, there is an element $v \in V^{**}$ which maps $f \mapsto f(v)$ for all $f \in V^*$.

To see that this is indeed an isomorphism (for finite dimensional vector spaces), choose a basis $\{v^i\}_{i \leq \dim V}$ for V , and it suffices to show that the corresponding elements of V^{**} are a basis. To specify an element of V^* , it suffices to specify it on the basis $\{v^i\}$. Define w_i as the map $v^j \mapsto \delta_j^i$. The set $\{w_i\}$ is linearly independent by the exact same argument offered in the previous section, so it forms a basis. Now define v^* as the map $w_j \mapsto \delta_j^i$. Once again, this is a basis. Now simply observe that v^* and v are the corresponding elements under the claimed isomorphism, proving it is indeed a bijection on bases. Linearity is immediate.

In particular, $W_p^i(\omega_p^i) = \omega_p^i(V_p^i)$, so we recover our set $\{V_p^i\}_{1 \leq i \leq n}$ with $\omega_p^i(V_p^j) = \delta_j^i$. From here, the proof proceeds as before. The set $\{\omega^i\}$ forms a smooth global frame by smoothness of Φ , so we may write covector fields η as

$$\eta_p = \sum_i c_i(p) \omega_p^i.$$

We may allow the above natural isomorphism to induce a smooth structure on $T^{**}M = (T^*M)^*$ such that it promotes to a diffeomorphism. Write the diffeomorphism as Ψ . Suppose we show the section W^i of $T^{**}M$ is smooth. Then since $W^i = \Psi \circ V^i$, we will have that $\Psi^{-1}W^i = V^i$ is smooth, so it suffices to show that the sections W^i of $T^{**}M$ are smooth. This process is exactly the same as before, with the only exception being that we must prove Lee 11.11 for the new co-cotangent bundle. We do this at the end. The c_i s are smooth since they arise as the smooth composite $\pi_{\mathbb{R}} \circ \Phi \circ \eta$ once again, and

$$W^i(\eta)_p = W^i \left(\sum_j c_j(p) \omega_p^j \right) = \sum_j c_j(p) \delta_j^i = c_i(p).$$

Therefore, each $W^i(\eta)$ is smooth since $W^i(\eta) = c_i$ is smooth.

Now we show this implies that W^i is smooth. Let $p \in M$ and $(U, x^1, \dots, x^n)$ a chart for M containing p . This induces a canonical frame ∂_i for TM , and thus a corresponding frame which we will denote D^i for $T^{**}M$. For all $q \in U$, we can write

$$W_q^i = \sum_j W_j^i(q) D^j(q).$$

Here we note that the notation $D^j(q)$ corresponds to $\partial_j|_q$ or $dx^j|_q$, and W_j^i is some function which defines W^i over U in these coordinates, as before. Then W^i is smooth if and only if the component functions W^i are all smooth. Fix some k ; we show W_k^i is smooth on U . Then

$$W_q^i(dx^k) = \sum_j W_j^i(q) D^j(dx^k) = \sum_j W_j^i \delta_k^j = W_k^i(q).$$

We know already that, $W_k^i(dx^k)$ is smooth since dx^k is a covector, so W_k^i is smooth. Therefore, W^i is smooth on any arbitrary chart U , and is thus globally smooth. ■

Problem 11.7

Problem 3.1. Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the smooth map $(s, t) \mapsto (st, e^t)$, and let $\omega \in \mathfrak{X}^*(\mathbb{R}^2)$ be given by $\omega = xdy - ydx$. Compute $F^*\omega$.

Proof. We will follow the notation in the problem, writing (s, t) for the coordinates in the domain and (x, y) for the coordinates in the codomain.

We will compute $F^*\omega$ by seeing how it acts on a tangent vector $V = V^s\partial_s|_p + V^t\partial_t|_p \in T_p\mathbb{R}^2$, where $p = (s, t) \in \mathbb{R}^2$. This is given by

$$(F^*\omega)_p(V) = \omega_{F(p)}(F_*pV).$$

To compute F_*pV , we compute the Jacobian of F at p :

$$J(F)|_p = \begin{pmatrix} \partial_s F^s & \partial_t F^s \\ \partial_s F^t & \partial_t F^t \end{pmatrix} = \begin{pmatrix} t & s \\ 0 & e^t \end{pmatrix}.$$

Then

$$F_*pV = \begin{pmatrix} t & s \\ 0 & e^t \end{pmatrix} \begin{pmatrix} V^s \\ V^t \end{pmatrix} = \begin{pmatrix} tV^s + sV^t \\ e^tV^t \end{pmatrix} = (tV^s + sV^t) \frac{\partial}{\partial x} \Big|_{F(p)} + e^tV^t \frac{\partial}{\partial y} \Big|_{F(p)}.$$

Note that $F(p) = (st, e^t) = (x, y)$. Acting ω on F_*pV yields by linearity and $dx^i(\partial_j) = \delta_i^j$ that

$$\begin{aligned} \omega_{F(p)}(F_*pV) &= (stdy - e^tdx) \left((tV^s + sV^t) \frac{\partial}{\partial x} \Big|_{F(p)} + e^tV^t \frac{\partial}{\partial y} \Big|_{F(p)} \right) \\ &= -e^t(tV^s + sV^t) + ste^tV^t = -te^tV^s + (ste^t - se^t)V^t. \end{aligned}$$

Therefore, $F^*\omega$ can be written as

$$F^*\omega = (t-1)se^tdt - te^tds.$$

■

Problem 11.14

Problem 4.1. Consider the covector fields on $\mathbb{R}^3$:

$$\omega = -\frac{4zdx}{(x^2+1)^2} + \frac{2ydy}{y^2+1} + \frac{2xdz}{x^2+1}$$

$$\eta = -\frac{4xzdxdx}{(x^2+1)^2} + \frac{2ydy}{y^2+1} + \frac{2dz}{x^2+1}.$$

1. Set up and evaluate the line integrals of each covector field along the line segment from 0 to $(1, 1, 1)$.
2. Determine whether or not these vector fields are exact.
3. For each one that is exact, find its potential function and use it to recompute the line integral.

Proof. (1) Write $I = [0, 1] \subseteq \mathbb{R}$ for the unit interval, and let $\gamma : I \rightarrow \mathbb{R}^3$ be the map $\gamma(t) = (t, t, t)$ parameterizing the given line segment. The line integrals are then given by

$$\int_{\gamma} \omega = \int_0^1 \gamma^*\omega, \quad \int_{\gamma} \eta = \int_0^1 \gamma^*\eta.$$

We start by computing the coordinate representations of $\gamma^*\omega$ and $\gamma^*\eta$. Let $V = V^t\partial_t$ be a tangent vector in T_tI , and consider $(\gamma^*\omega)_t(V) = \omega_{(t,t,t)}(\gamma_*V)$. The Jacobian of γ is

$$J(\gamma) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Thus

$$\gamma_*V = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} V^t = \begin{pmatrix} V^t \\ V^t \\ V^t \end{pmatrix}.$$

Then we have

$$\omega_{(t,t,t)}(\gamma_*V) = \begin{pmatrix} -\frac{4t}{(t^2+1)^2} & \frac{2t}{t^2+1} & \frac{2t}{t^2+1} \end{pmatrix} \begin{pmatrix} V^t \\ V^t \\ V^t \end{pmatrix} = V^t \left(\frac{4t}{t^2+1} - \frac{4t}{(t^2+1)^2} \right).$$

We then obtain that

$$\gamma^*\omega = \left(\frac{4t}{t^2+1} - \frac{4t}{(t^2+1)^2} \right) dt.$$

We can compute the integral of this.

$$\int_{\gamma} \omega = \int_0^1 \gamma^*\omega = \int_0^1 \frac{4t}{t^2+1} - \frac{4t}{(t^2+1)^2} dt = \int_0^1 \frac{4t}{t^2+1} dt - \int_0^1 \frac{4t}{(t^2+1)^2} dt.$$

For both, use the u -substitution $u = t^2 + 1$. This yields $du = \frac{\partial u}{\partial t} dt = 2t$, and $u(0) = 1$ and $u(1) = 2$, so the bounds of the integrals do not change. Completing the computation:

$$\int_1^2 \frac{2}{u} du - \int_1^2 \frac{2}{u^2} du = 2 \left[\ln u + \frac{1}{u} \right]_1^2 = 2 \left(\ln 2 + \frac{1}{2} - 1 \right) = 2 \ln 2 - 1.$$

Now we compute the integral of η . We have

$$\begin{aligned} (\gamma^*\eta)_p(V) &= \eta_{(t,t,t)}(\gamma_*V) = \begin{pmatrix} -\frac{4t^2}{(t^2+1)^2} & \frac{2t}{t^2+1} & \frac{2}{t^2+1} \end{pmatrix} \begin{pmatrix} V^t \\ V^t \\ V^t \end{pmatrix} \\ &= V^t \left(\frac{2t+2}{t^2+1} - \frac{4t^2}{(t^2+1)^2} \right). \end{aligned}$$

We thus have that in coordinates,

$$\gamma^*\eta = \left(\frac{2t+2}{t^2+1} - \frac{4t^2}{(t^2+1)^2} \right) dt.$$

The integral is then

$$\int_{\gamma} \eta = \int_0^1 \gamma^*\eta = \int_0^1 \frac{2t+2}{t^2+1} - \frac{4t^2}{(t^2+1)^2} dt = \int_0^1 \frac{2t}{t^2+1} dt + \int_0^1 \frac{2}{t^2+1} dt - \int_0^1 \frac{4t^2}{(t^2+1)^2} dt.$$

We already saw the integral of the first integral is $\ln u|_1^2 = \ln 2$. We recognize the integrand of the second integral as the derivative of arctan, so that evaluates to

$$2(\arctan 1 - \arctan 0) = 2\left(\frac{\pi}{4} - 0\right) = \frac{\pi}{2}.$$

To compute the final integral, we ask chatgpt for the correct substitution $t = \tan \theta$. Then $dt = \frac{\partial t}{\partial \theta} d\theta = \sec^2 \theta d\theta$. Note that $\tan \frac{\pi}{4} = 1$ and $\tan 0 = 0$, so the integral becomes

$$- \int_0^{\pi/4} \frac{4 \tan^2 \theta}{(\tan^2 \theta + 1)^2} \sec^2 \theta d\theta.$$

Note that

$$\tan^2 \theta + 1 = \frac{\sin^2 \theta}{\cos^2 \theta} + 1 = \frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}.$$

Therefore

$$\frac{4 \tan^2 \theta \sec^2 \theta}{(\tan^2 \theta + 1)^2} = \frac{4 \tan^2 \theta \sec^2 \theta}{\frac{1}{\cos^4 \theta}} = 4 \tan^2 \theta \sec^2 \theta \cos^4 \theta = 4 \frac{\sin^2 \theta}{\cos^2 \theta} \frac{1}{\cos^2 \theta} \cos^4 \theta = 4 \sin^2 \theta.$$

The integral is thus

$$- \int_0^{\pi/4} 4 \sin^2 \theta d\theta.$$

Use the double angle identity

$$\cos 2\theta = 1 - 2 \sin^2 \theta \implies \sin^2 \theta = \frac{1 - \cos 2\theta}{2}.$$

This yields

$$= -2 \int_0^{\pi/4} 1 - \cos 2\theta d\theta = -2 \left(\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} + \frac{1}{2} \sin 0 \right) = -2 \left(\frac{\pi}{4} - \frac{1}{2} \right) = 1 - \frac{\pi}{2}.$$

The full integral is thus

$$2 \ln 2 + \frac{1}{2} - \frac{\pi}{2} + \frac{\pi}{2} = 2 \ln 2 + 1.$$

(2) Since $M = \mathbb{R}^3$, we can use calculus 3 knowledge. Note that the gradient df in coordinates is precisely the usual gradient:

$$df = \sum_i \frac{\partial f}{\partial x^i} dx^i = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = \nabla f.$$

Of course we are replacing the basis vectors $\hat{x}, \hat{y}, \hat{z}$ with dx^i , but this is an isomorphism of vector spaces so is negligible. A vector field in $\mathbb{R}^3$ is exact if and only if it is conservative if and only if¹ it has 0 curl.

$$\begin{aligned} \nabla \times \omega &= (\partial_y \omega_z - \partial_z \omega_y) dx - (\partial_x \omega_z - \partial_z \omega_x) dy + (\partial_x \omega_y - \partial_y \omega_x) dz \\ &= - \left(\frac{2x^2 + 2 - 4x^2}{(x^2 + 1)^2} + \frac{4}{(x^2 + 1)^2} \right) dy = \frac{2x^2 - 6}{(x^2 + 1)^2} dy \neq 0. \end{aligned}$$

Therefore, ω is not exact. We now compute the curl of η .

$$\nabla \times \eta = - \left(-\frac{4x}{(x^2 + 1)^2} + \frac{4x}{(x^2 + 1)^2} \right) dy = 0.$$

¹I actually don't remember if this second statement is if and only if, but by problem 11.15 we show that if its conservative it has 0 gradient, so by contrapositive ω is not conservative. By construction of a potential in part (3), η is conservative. So if this assumption fails, the proof still works.

Therefore η is exact.

(3) I claim that $\eta = df$ when

$$f = \frac{2z}{x^2 + 1} + \ln(y^2 + 1).$$

Indeed,

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = -\frac{4xz}{(x^2 + 1)^2} dx + \frac{2y}{y^2 + 1} dy + \frac{2}{x^2 + 1} dz = \eta.$$

We then have by the fundamental theorem for line integrals that

$$\int_{\gamma} \eta = \int_{\gamma} df = f(\gamma(1)) - f(\gamma(0)) = f(1, 1, 1) - f(0, 0, 0) = \frac{2}{2} + \ln 2 = 1 + \ln 2.$$

This is precisely the same answer we got before, but with a lot less work. ■

Problem 11.15

Problem 5.1. Let X be a smooth vector field defined on some open $U \subseteq \mathbb{R}^n$. Given a piecewise smooth $\gamma : [a, b] \rightarrow U$, define the line integral of X over γ , denoted $\int_{\gamma} X \cdot ds$, as

$$\int_{\gamma} X \cdot ds = \int_a^b X_{\gamma(t)} \cdot \gamma'(t) dt$$

where $\cdot$ is the usual dot product. A conservative vector field is one whose line integral around every piecewise smooth closed curve is 0.

1. Show X is conservative if and only if there exists $f \in C^\infty(U)$ such that $X = \nabla f$ (viewing ∇f as a section of TU). [Hint: consider $\omega \in \mathfrak{X}^*(U)$ given by $\omega_x(v) = X_x \cdot v$]
2. Suppose $n = 3$, and define curl in the usual way. Show that if X is conservative, then $\nabla \times X = 0$.
3. Show that if $U \subseteq \mathbb{R}^3$ is star-shaped, then X is conservative on U if and only if $\nabla \times X = 0$.

Proof. We start by constructing an integral-preserving isomorphism $TU \cong T^*U$ using the global coordinates on U induced by $\mathbb{R}^n$. Let $\varphi : TU \rightarrow T^*U$ be the map $\partial_i \mapsto dx^i$, where we use the canonical coordinates in $\mathbb{R}^n$. This induces a map $\mathfrak{X}(U) \rightarrow \mathfrak{X}^*(U)$ defined by

$$X = \sum_i X^i \partial_i \mapsto \varphi(X) = \sum_i X^i dx^i.$$

This map is either a diffeomorphism $TM \cong T^*M$ by obviousness, by inspection of the coordinates, or by problem 11.5. This fact is not useful beyond the fact that the induced map $\mathfrak{X}(U) \rightarrow \mathfrak{X}^*(U)$ is a bijection, which is also clear from the fact that it is a bijection of basis elements on each tangent space $T_p M \cong T_p^* M$. We claim that φ , viewed as a map $\mathfrak{X}(U) \cong \mathfrak{X}^*(U)$, preserves integrals in the sense that using the given definition,

$$\int_{\gamma} X \cdot ds = \int_{\gamma} \varphi(X)$$

for any piecewise smooth $\gamma : [a, b] \rightarrow U$. By Lee 11.38,

$$\int_{\gamma} \varphi(X) = \int_a^b \varphi(X)_{\gamma(t)}(\gamma'(t)) dt.$$

By definition,

$$\int_{\gamma} X \cdot ds = \int_a^b X_{\gamma(t)} \cdot \gamma'(t) dt.$$

We must show these are the same. Indeed, writing γ^i for the i th component of γ and recalling how velocity vectors are written in coordinates,

$$\begin{aligned} \varphi(X)_{\gamma(t)}(\gamma'(t)) &= \sum_i X_{\gamma(t)}^i dx^i(\gamma'(t))|_{\gamma(t)} = \sum_i X_{\gamma(t)}^i dx^i \left(\sum_j \frac{d\gamma^j}{dt} \Big|_t \frac{\partial}{\partial x^j} \Big|_{\gamma(t)} \right) \Big|_{\gamma(t)} \\ &= \sum_i X_{\gamma(t)}^i \frac{d\gamma^i}{dt} = X_{\gamma(t)} \cdot \gamma'(t). \end{aligned}$$

This is precisely the integrand of the second integral, completing the proof. We now complete the problem.

(1) By the above argument, φ and its inverse φ^{-1} preserve conservativeness. It suffices to show that $\varphi(\nabla f) = df$, since the converse will then follow for free:

$$\varphi^{-1}(df) = (\varphi^{-1} \circ \varphi)(\nabla f) = \nabla f.$$

Indeed, in coordinates,

$$\nabla f = \sum_i \frac{\partial f}{\partial x^i} \frac{\partial}{\partial x^i},$$

and

$$df = \sum_i \frac{\partial f}{\partial x^i} dx^i.$$

We thus have

$$\varphi(\nabla f) = \sum_i \frac{\partial f}{\partial x^i} dx^i = df.$$

(2) This is a calculus 3 proof which we recite. Let $X = \nabla f$ be a vector field on U . Then

$$\nabla \times X = \nabla \times \nabla f = \left(\frac{\partial}{\partial y} \frac{\partial f}{\partial z} - \frac{\partial}{\partial z} \frac{\partial f}{\partial y} \right) \frac{\partial}{\partial x} - \left(\frac{\partial}{\partial x} \frac{\partial f}{\partial z} - \frac{\partial}{\partial z} \frac{\partial f}{\partial x} \right) \frac{\partial}{\partial y} + \left(\frac{\partial}{\partial x} \frac{\partial f}{\partial y} - \frac{\partial}{\partial y} \frac{\partial f}{\partial x} \right) \frac{\partial}{\partial z}.$$

Since f is smooth, the partials commute with each other, so we can rewrite this as

$$\nabla \times \nabla f = \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial y \partial z} \right) \frac{\partial}{\partial x} - \left(\frac{\partial^2 f}{\partial x \partial z} - \frac{\partial^2 f}{\partial x \partial z} \right) \frac{\partial}{\partial y} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial x \partial y} \right) \frac{\partial}{\partial z} = 0.$$

(3) The condition $\nabla \times f = 0$ is equivalent to saying that

$$\frac{\partial^2 f}{\partial x^i \partial x^j} = \frac{\partial^2 f}{\partial x^j \partial x^i},$$

and it follows by the chain rule that this is true in any chart (V, ψ) . Translating under φ into the language of covector fields, is the definition of a closed vector field. Therefore, closed vector fields are precisely those whose curl vanishes. By Lee proposition 11.44 and Poincaré's lemma (Lee 11.49), closed covector fields are precisely exact covector fields on star-shaped domains, and thus the same is true for vector fields. ■